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2022高数下期末.

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一、

(1) $\lim_{n\to\infty}\left|\frac{1}{n^n\cdot 3}\right| = \frac{1}{3}$. $x\in[-3,3)$, D

(2) $\left.\frac{\partial f}{\partial x}\right|_{(0,0)} = \lim_{x\to 0}\frac{0-0}{x} = 0$, $\left.\frac{\partial f}{\partial y}\right|_{(0,0)} = \lim_{y\to 0}\frac{y^2\sin\frac{1}{y^2}-0}{y} = \lim_{y\to 0} y\sin\frac{1}{y^2} = 0$,

$\frac{\partial f}{\partial x} = y^2\cdot\frac{-2x}{(x^2+y^2)^2}\cdot\cos\left(\frac{1}{x^2+y^2}\right)$. $\lim_{(x,y)\to(0,0)}\frac{\partial f}{\partial x}\neq 0$, 不连续 D

(3). C.

(4). B

(5) $\begin{cases} 0\le y\le 1 \\ y\le x\le 1 \end{cases} \Rightarrow 0\le y\le x\le 1 \Rightarrow \int_0^1\int_0^x \frac{\cos x}{x}\,dy\,dx$

$= \int_0^1 \cos x\,dx = \sin 1$. B.

二、

(1) 先代入 $x=5$, 得: $z^2+2z-2y^2=0$.

$\frac{\partial z}{\partial y} = \frac{-F_y}{F_z} = \frac{4y}{2z+2} = \frac{4}{3}$

(2). 原式 $= \lim_{(x,y)\to(0,0)}\frac{xy^2}{x^2+y^2} \le \lim_{(x,y)\to(0,0)} x = 0$.

又, 原式 >0, 故原式 $=0$

(3) $= \int_0^{2\pi}\int_0^1 e^{-r^2}\cdot r\,dr\,d\theta = \int_0^{2\pi}\frac{1}{2}(1-\frac{1}{e})\,d\theta = \pi(1-\frac{1}{e})$

(4) $\frac{2}{3}a^3\pi$

(5). 保守场: $\frac{\partial f}{\partial x} = yze^{xz}$. $f = ye^{xz}+g(y,z)$, $\frac{\partial f}{\partial y} = e^{xz}+\frac{\partial g}{\partial y}$,

$\frac{\partial g}{\partial y} = 0 \Rightarrow f = ye^{xz}+h(z)$, $\frac{\partial f}{\partial z} = xye^{xz}+h'(z)$, $h(z)=C$, $\Rightarrow f = ye^{xz}+C$

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代入，得：$4\cdot e^0-1\cdot e^0=3$.

三、 l_1方向向量：$\vec{v_1}=\langle 0,1,1\rangle$；$l_2$方向向量 $\vec{v_2}=\langle 1,2,1\rangle$

$$\vec{n}=\vec{v_1}\times\vec{v_2}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\0&1&1\\1&2&1\end{vmatrix}=\hat{i}+\hat{j}-\hat{k}-2\hat{i}=\langle -1,1,-1\rangle$$

$\Rightarrow$ 平面方程：$-x+y-z=0$.

四、 原式 $=\lim\limits_{n\to\infty} n^3\left(\frac{2}{n}-\frac{1}{3!}\cdot\left(\frac{2}{n}\right)^3+\cdots\right)-2n^2$

$=\lim\limits_{n\to\infty} 2n^2-\frac{8}{3!}-2n^2=-\frac{4}{3}$

五、 $\nabla f=(y)\hat{i}+(x+2y)\hat{j}$ $\quad \nabla f|_{(3,2)}=2\hat{i}+7\hat{j}$

又有且仅有$\langle -7,2\rangle$或$\langle 7,-2\rangle$所在方向的向量与$\nabla f|_{(3,2)}$内积为0，

故：方向1：$\vec{v_1}=\left\langle\frac{-7}{\sqrt{53}},\frac{2}{\sqrt{53}}\right\rangle$所在方向.

方向2：$\vec{v_2}=\left\langle\frac{7}{\sqrt{53}},\frac{-2}{\sqrt{53}}\right\rangle$所在方向.

六、 $D:(x-1)^2+(y-1)^2=2$ 令 $v=x-1$，$u=y-1$，$J=\left|\frac{\partial(x,y)}{\partial(v,u)}\right|=1$.

$$\text{原式}=\iint_D (v+1)(u+1)\,dA=\int_0^{2\pi}\int_0^{\sqrt{2}}(r^2\sin\theta\cos\theta+r\sin\theta+r\cos\theta+1)r\,dr\,d\theta$$

$$=\int_0^{2\pi}\left[\frac{1}{4}r^4\sin\theta\cos\theta+\frac{1}{3}r^3(\sin\theta+\cos\theta)+\frac{1}{2}r^2\right]_0^{\sqrt{2}}d\theta$$

$$=\int_0^{2\pi}d\theta=2\pi$$

七、 令 $\sqrt{x^2+y^2}=\sqrt{1-x^2-y^2}$，$\Rightarrow x^2+y^2=\frac{1}{2}$

$$M=\iiint_R dV=\int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}}\int_r^{\sqrt{1-r^2}} r\,dz\,dr\,d\theta=\int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}} r\sqrt{1-r^2}-r^2\,dr\,d\theta$$

$$=\int_0^{2\pi}\left[-\frac{1}{3}(1-r^2)^{\frac{3}{2}}-\frac{1}{3}r^3\right]_0^{\frac{1}{\sqrt{2}}}d\theta=\int_0^{2\pi}\frac{1}{3}-\frac{1}{3\sqrt{2}}\,d\theta=\left(\frac{1}{3}-\frac{1}{3\sqrt{2}}\right)2\pi$$

$$M_{xy}=\int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}}\int_r^{\sqrt{1-r^2}} rz\,dz\,dr\,d\theta=\int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}}\frac{1}{2}r(1-2r^2)\,dr\,d\theta$$

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$= \int_0^{2\pi} \left[\frac{1}{4}r^2 - \frac{1}{4}r^4\right]_0^{\frac{1}{\sqrt{2}}} d\theta = \int_0^{2\pi} \frac{1}{16} d\theta = \frac{\pi}{8}$

$\Rightarrow \bar{z} = \frac{9\sqrt{2}}{48(\sqrt{2}-1)}$

由对称性，得 $\bar{x} = \bar{y} = 0$.

故：形心为 $(0, 0, \frac{9\sqrt{2}}{48(\sqrt{2}-1)})$

八、$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \nabla \times \vec{F} \cdot \vec{n}\, dS = 0$

九、(i) 在 $x^2+y^2=1$ 上.

令 $g(x,y) = x^2+y^2-1 = 0$,

$\nabla f = \lambda \nabla g$, $\Rightarrow (6x+4y)\hat{i} + 4x\hat{j} = \lambda[(2x)\hat{i} + 2y\hat{j}]$

$\Rightarrow \begin{cases} 6x+4y = 2\lambda x \\ 4x = 2\lambda y \end{cases} \Rightarrow 3\lambda + 4 = \lambda^2$, $\lambda = 4$ 或 $\lambda = -1$

① $\lambda = 4$, $\Rightarrow y = \pm\frac{1}{\sqrt{5}}$, $x = \pm\frac{2}{\sqrt{5}}$

② $\lambda = -1$, $\Rightarrow y = \pm\frac{2}{\sqrt{5}}$, $x = \mp\frac{1}{\sqrt{5}}$

取 $\begin{cases} y = \frac{1}{\sqrt{5}} \\ x = \frac{2}{\sqrt{5}} \end{cases}$ 代入, $f = \frac{12}{5} + \frac{8}{5} = 4$

取 $\begin{cases} y = \frac{2}{\sqrt{5}} \\ x = -\frac{1}{\sqrt{5}} \end{cases}$ 代入, $f = -1$

(ii) $\frac{\partial f}{\partial x} = 6x+4y$, $\frac{\partial f}{\partial y} = 4x$. 当 $f_x = f_y = 0$ 时: $x = y = 0$.

此时 $f = 0$ $f_{xx} = 6 > 0$,

$f_{yy} = 0$, $f_{xy} = 4$, $f_{xx}f_{yy} - f_{xy}^2 < 0$, $\Rightarrow$ 是 saddle point.

综上所述，最大值在 $(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}})$ 和 $(\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}})$ 处取得，为 4，最小值在 $(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}})$ 和 $(\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}})$ 处取得，为 -1.